

Expected multi-utility representation $\Rightarrow$ Independence

Result — ai; produced by Claude Fable 5.1 (Cowork session).

0.1 Premises

- **Expected multi-utility representation.** There is a set U of functions $u : X \rightarrow \mathbb{R}$ such that for all $p, q \in \Delta(X)$: $p \succcurlyeq q$ if and only if $EU_u(p) \geq EU_u(q)$ for every $u \in U$.

0.2 Conclusion

- **Independence.** For all $p, q, r \in \Delta(X)$ and $\lambda \in (0,1]$: $p \succcurlyeq q$ if and only if $\lambda p + (1-\lambda)r \succcurlyeq \lambda q + (1-\lambda)r$.

0.3 Proof

EU_u is affine in mixtures, so for $\lambda \in (0,1]$: $EU_u(\lambda p + (1-\lambda)r) - EU_u(\lambda q + (1-\lambda)r) = \lambda(EU_u(p) - EU_u(q))$, which is ≥ 0 iff $EU_u(p) \geq EU_u(q)$. Quantify over $u \in U$.

Record: topics/decision-theory/results/multi-utility-implies-independence.yaml